Activity 6: Yield and Performance Metric Verification

Course: Semiconductor Device and Process Integration Masterclass

Course code: C605

Duration: 55 minutes

PPT alignment: Trainer Slides v1.0, concept slides 298-355 and Activity 6 overview slide 421.

Workplace scenario

A wafer-sort yield decline is concentrated near the edge while centre-die parametric results remain within specification.

Goal

Select and interpret metrics that verify process-integration performance.

Files in this folder

- README.md / README.pdf - complete procedure and acceptance criteria.
- SCENARIO.md / SCENARIO.pdf - facts, assumptions and role brief.
- WORKSHEET.md / WORKSHEET.pdf - structured learner response.
- EVIDENCE-CHECKLIST.md / EVIDENCE-CHECKLIST.pdf - completion and review record.
- Any CSV file in this folder - supplied teaching data for analysis.

Step-by-step procedure

1. Inspect the supplied lot-level yield, wafer-map, CD, film-thickness and sheet-resistance data.
2. Calculate overall yield, edge-versus-centre yield, mean, range and the specified capability proxy.
3. Separate product-performance metrics from process-stability, yield and reliability metrics.
4. Test whether the spatial signature is consistent with coating, plasma, thermal or handling mechanisms.
5. State a verification conclusion, confidence level and the next measurement needed before disposition.

Required evidence

Metric dashboard worksheet with calculations, spatial interpretation and verification conclusion.

Include your completed worksheet, any calculations or annotated diagrams, the evidence checklist, and a short note identifying assumptions or unresolved evidence gaps.

Acceptance criteria

[ ] All required metrics are calculated correctly

[ ] The wafer-map pattern informs the conclusion

[ ] Conclusion distinguishes verification from speculation

Verification

Before submission, ask a peer to challenge one causal link. Revise the conclusion if the challenge reveals that evidence and inference have been mixed. Your final response must distinguish: observed fact, interpretation, decision, and follow-up check.

Troubleshooting

- If several mechanisms fit the symptom, choose a measurement that discriminates between them.
- If a process module lacks a defined input, output or control gate, the flow is incomplete.
- If an action is not tied to a verified cause, record it as a hypothesis test rather than corrective action.
- If calculations disagree, show the method and units so the discrepancy can be reviewed.